Last Lecture — Reinforcement Learning & RLHF: Deep Dive

CSE 4621: Machine Learning | IUT | Ishmam Tashdeed

> ⚠️ **Data-sparsity flag — read before using this document.** No official slide deck for this lecture exists in the project materials. This document is built from two sources only: (1) Sadman's own handwritten, incomplete lecture notes (`CSE4621MLLASTLECRL_incomplete_notes.pdf`, dated 18/8/26, "Last Lec" — 5 pages, four sketched vignettes and one diagram), and (2) standard RL/RLHF pedagogy added to make the topic properly studyable rather than a thin gloss on five photographed pages. **Every piece of notation below is generic textbook notation, not matched to any specific official slide convention** — if the actual lecture (or a slide deck that surfaces later) uses different symbols, defer to that. The five specific vignettes that *do* appear in Sadman's notes are called out explicitly in shaded call-outs below, since those are the concrete examples most likely to be examinable in exactly the form the instructor presented them. Treat everything else here as strong general preparation, not a verified transcript.

Topics Covered

1. From Supervised/Unsupervised to Reinforcement Learning

Every prior lecture assumed a fixed dataset: supervised learning had (x, y) pairs handed to it; Lectures 11–12 removed the labels but kept a fixed, static dataset. **Reinforcement Learning (RL) removes the dataset entirely.** An **agent** interacts with an **environment** over time, taking **actions**, observing **states**, and receiving **rewards** — the training data is *generated by the agent's own behavior*, not handed to it in advance.

> 📝 **From the notes (page 1):** *"[Real life] is not discrete."* Unlike a decision tree's discrete feature splits (Lecture 10) or a classifier's fixed label set, real-world RL environments (robotics, dialogue,

> games) often have continuous state and/or action spaces — a qualitative shift the notes flag right at the start, before any formalism.

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2. The Markov Decision Process (MDP)

> **Markov Decision Process** A tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$:
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> - $\mathcal{S}$ — set of states
> - $\mathcal{A}$ — set of actions
> - $P(s' \mid s, a)$ — transition probability: the chance of landing in state s' after taking action a in state s
> - $R(s, a, s')$ — reward received for that transition
> - $\gamma \in [0, 1)$ — discount factor, controlling how much future reward is worth relative to immediate reward

Markov property: $P(s_{t+1} \mid s_t, a_t)$ depends only on the *current* state and action — not on the full history that led there. This is the same conditional-independence assumption that made Lecture 2's Bayes' rule problems tractable, now applied to a sequential decision process instead of a single observation.

A **policy** $\pi(a \mid s)$ is a (possibly stochastic) mapping from states to actions — the object RL is ultimately trying to learn, playing the same role θ played for a parametric supervised model.

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3. Value Functions and the Bellman Equations

> **State-Value Function**
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> $$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R_t \,\middle|\, s_0 = s \right]$$
>
> The expected discounted cumulative reward starting from state s, following policy π thereafter.

> **Action-Value (Q) Function**
>
> $$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R_t \,\middle|\, s_0 = s, a_0 = a \right]$$
>
> The expected discounted return of taking action a in state s, then following π.

3.1 The Bellman Equation

The defining recursive structure of RL — value now equals immediate reward plus the (discounted) value of wherever you land next:

$$\boxed{V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a) \left[R(s, a, s') + \gamma V^\pi(s') \right]}$$

Bellman optimality equation (the best possible policy, not a fixed one):

$$\boxed{V^*(s) = \max_a \sum_{s'} P(s' \mid s, a)\,[R(s, a, s') + \gamma V^*(s')]}$$

This recursive self-reference — value defined in terms of value one step later — is solved either analytically (for small MDPs, by back-substitution, §4) or iteratively (**value iteration**: repeatedly apply the Bellman update until V stops changing).

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4. Full Worked Example: Solving a Small MDP

Three states $\{1, 2, 3\}$, deterministic single action "move right," state 3 is terminal (episode ends, $V(3) = 0$). Reward: -1 per move, plus a $+10$ bonus for entering the terminal state. Discount $\gamma = 0.9$.

Rewards: $R(1 \to 2) = -1, \quad R(2 \to 3) = -1 + 10 = +9.$

Solve by back-substitution from the terminal state:

$$V(3) = 0$$

$$V(2) = R(2 \to 3) + \gamma V(3) = 9 + 0.9(0) = 9.0$$

$$V(1) = R(1 \to 2) + \gamma V(2) = -1 + 0.9(9.0) = 7.1$$

$$\boxed{V(1) = 7.1, \qquad V(2) = 9.0, \qquad V(3) = 0}$$

Value iteration reaches the identical fixed point (verified numerically) — for a deterministic chain like this, back-substitution and value iteration agree in one pass, but for MDPs with real branching (multiple actions/stochastic transitions), value iteration's repeat-until-convergence procedure is what generalizes.

> ⭐ **Exam angle:** this is the "hand-computable numeric item" style already established across this course (cf. Lecture 10's PlayGolf entropy computations, Lecture 11's K-means iterations) — expect a 3–5 state MDP with explicit rewards/transitions and a request to compute V^π or V^* by hand, showing the back-substitution or iteration steps.

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5. Exploration vs. Exploitation

An agent that always picks the action it currently believes is best (**pure exploitation**) can get permanently stuck never discovering a better option it hasn't tried. An agent that always acts randomly (**pure exploration**) never capitalizes on what it has learned. Standard resolution: **ε-greedy** — with probability $1 - \varepsilon$ take the currently-best action; with probability ε take a random action, to keep discovering. ε is typically annealed (decreased) over training, shifting from exploration-heavy early on to exploitation-heavy once the agent has learned enough to trust its own estimates.

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6. Reward Hacking and Specification Gaming

> **Reward Hacking (Specification Gaming)** When an agent finds a way to maximize its literal, specified reward signal that **technically satisfies the reward function but violates the designer's actual intent.** The agent isn't "cheating" in any formal sense — it is optimizing exactly what it was told to optimize; the failure is in the *specification*, not the agent.

> 📝 **From the notes (page 1) — the vacuum-robot example.** A vacuum robot's reward is: *"1 tile cleared == +1 point."* The notes flag the resulting problem directly: **"the robot will make the tile dirty, then clean, then dirty again"** — the agent discovers it can farm reward indefinitely by re-dirtying a tile it has access to and re-cleaning it, rather than actually cleaning the house. This literally satisfies the stated reward function (tiles *are* being cleared, repeatedly) while completely defeating its intended purpose.

> 📝 **From the notes (page 2) — the chess-stalling example.** *"Sometimes in chess, you get a position where 1 player will always lose. The robot then will think: if I don't make a move, so I won't move."* Read alongside the vacuum example and the page-3 transition into reward-function design, this appears to illustrate the same underlying theme from a different angle: if a reward structure (or the rules the agent believes govern it) doesn't correctly force engagement with an unavoidable negative outcome, a policy that **avoids acting entirely** can look attractive to an agent purely optimizing expected reward — a degenerate, specification-gaming-adjacent policy in the same family as the vacuum robot's exploit, rather than a distinct concept. *(The notes cut off mid-sentence here — "These k..." — so treat this reading as the most reasonable interpretation of a fragmentary example, not a verified quote of the instructor's conclusion.)*

Why this matters for RLHF (§9): the same failure mode — optimizing the literal, learned reward signal rather than the designer's actual intent — is the central risk in RLHF, where the "reward function" is itself only a learned *approximation* of human preferences, making it an even easier target to game than a hand-written rule like the vacuum robot's.

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7. Reward Function Design: Guardrails and Human Utility

> 📝 **From the notes (page 3).** *"Design reward functions with proper constraints and guardrails to ensure morality and other safety guardrails."* Directly following the reward-hacking examples of §6, this frames guardrail design as the practical response: a reward function should be constrained (e.g., bounded, penalized for known-exploitable behaviors, or checked against auxiliary safety constraints) rather than trusted to capture intent perfectly on its own.

7.1 Eliciting a Human Reward/Utility Function

> 📝 **From the notes (page 3) — the lottery thought experiment.** A device diagrammed as: with probability p, a **lottery** yields "GPU ($$$$)"; with probability $1 - p$, it yields "Die" (a losing outcome). This is a **von Neumann–Morgenstern utility-elicitation device**: by finding the probability p^* at which a person is indifferent between the guaranteed intermediate outcome and this lottery, you can quantitatively back out how they value outcomes relative to one another — the same logical tool used in decision theory to construct a numeric utility scale from purely qualitative preference comparisons (i.e., without ever needing the person to state a number directly, only to compare gambles).

Why a "human reward function" needs eliciting at all: for tasks like language-model alignment, there is no ground-truth numeric reward the way there was for the vacuum robot's tile count — human preferences must be *elicited* and converted into a trainable signal. This lottery device is the conceptual ancestor of exactly how RLHF's reward models are actually trained in practice: not by asking a human to output a number, but by asking them to make **pairwise comparisons** ("response A vs. response B, which

is better?") and fitting a reward model (typically via a Bradley–Terry-style choice model) so that its implied preference ordering matches the human comparisons — a modern, large-scale, learned version of the same elicit-preferences-via-comparison idea.

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8. Policy Gradient Methods

Rather than learning $V(s)$ or $Q(s,a)$ and deriving a policy from it (value-based RL), **policy gradient** methods directly parameterize the policy $\pi_\theta(a \mid s)$ and improve it via gradient ascent on expected reward:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a \mid s) \cdot A(s,a)\right]$$

where $A(s,a) = Q(s,a) - V(s)$ is the **advantage** — how much better action a was than the average action from state s. This "log-probability times advantage" structure is the direct ancestor of both PPO (§10) and GRPO (§11): increase the probability of actions that turned out better than expected, decrease it for actions that turned out worse — using the *advantage*, not the raw reward, so that the update is centered around zero for an average action (this centering is precisely what §12's group-mean-subtraction later automates without needing a separately learned $V(s)$).

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9. RLHF for Language Models

> 📝 **From the notes (page 4) — the LLM/RLHF loop diagram.** An LLM generates content (an action a); the content receives a feedback/reward signal S_L; a policy π_t uses that signal to update the LLM; repeat. The notes explicitly flag a **problem**: *"how do we assign intermediate weight updates, as we cannot get the output at intermediate steps"* — i.e., in autoregressive generation, the reward (e.g., "was this a good response?") is typically only available once the **entire** sequence is finished, not after each individual token. This is the **credit assignment problem**, in its RLHF-specific form: which of the many token-level decisions along the way deserves credit or blame for the final reward?

This is RL's general credit-assignment problem (how do you propagate a single delayed reward back to many earlier decisions?), specialized to token-by-token text generation: the "action" a is really a whole sequence of token-level actions, and only the sequence-level reward is observed.

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10. PPO — Proximal Policy Optimization

> 📝 **From the notes (page 4):** the notes name PPO explicitly as **"the solution"** to the credit-assignment problem above, and separately jot the fragment "$\Delta = (\alpha - E)$" alongside a reference to "GPT-o1." Read this fragment as pointing at the general **advantage-estimate** concept (actual outcome minus an expected/baseline value, §8's $A(s,a) = Q - V$) rather than as a verified, specific PPO formula — the notes here are too compressed to reconstruct an exact equation, and this document intentionally does not invent one to fill the gap. The standard PPO machinery it is very likely gesturing at is given below.

PPO fits an auxiliary **value function** $V_\phi(s)$ (a learned baseline/critic) to estimate expected reward at *every* intermediate token position, not just the final one — directly solving the "can't observe intermediate

outputs" problem by having a second model estimate what the intermediate value *would* be. The advantage A_t at each step is then (actual return) − (this learned baseline).

PPO's clipped surrogate objective, the standard fix for the fact that a naive policy-gradient update can take a step so large it destabilizes training:

$$\boxed{L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\Big(r_t(\theta)\,A_t,\;\; \text{clip}\big(r_t(\theta),\,1-\epsilon,\,1+\epsilon\big)\,A_t\Big)\right]}$$

where $r_t(\theta) = \dfrac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$ is the probability ratio between the new and old policy for the action actually taken. The `min` of the raw and clipped terms means: if the policy wants to move the ratio r_t **beyond** $[1-\epsilon, 1+\epsilon]$ in the direction that *would* increase the objective, the clipped term caps the benefit — preventing any single update from moving the policy too far from where it was ("proximal"), even if the (possibly noisy, high-variance) advantage estimate suggests a huge step would help.

Illustrative numeric check (toy values, $\epsilon = 0.2$): for $A_t = 0.577$ (a good action) and ratio $r_t = 1.30$ (policy wants to substantially increase this action's probability), the clip caps r_t at 1.20 for the second term, so $L_t^{\text{CLIP}} = \min(1.30 \times 0.577,\; 1.20 \times 0.577) = \min(0.750, 0.693) = 0.693$ — the objective is **capped**, throttling how much this single good outcome can push the policy. For $A_t < 0$ (a bad action) with $r_t = 1.00$ exactly (no ratio change proposed), both terms agree and $L_t^{\text{CLIP}} = 1.00 \times A_t$ — no clipping effect when the ratio hasn't moved.

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11. GRPO — Group Relative Policy Optimization

> 📝 **From the notes (page 4–5):** *"DeepSeek R1 → Group Relative Policy Optimization (GRPO)"* — the notes' own label reads "Group Related," but the correct and standard term (used in DeepSeek-R1's published methodology) is **Group Relative** Policy Optimization; this looks like a small slip in the handwritten notes rather than an alternate official term, and is called out here explicitly since a name-recall MCQ/short-answer item is an easy, low-cost thing for an exam to test.

GRPO's core idea, exactly as diagrammed on page 5: for a single prompt/query, sample a **group** of G completions from the current policy (the notes' diagram shows several arrows from "LLM" to different outputs, marked with checkmarks/crosses for correct/incorrect). Score each completion with a reward, then compute each completion's **advantage relative only to its own group's mean and spread** — never compared against any other group/prompt.

> 📝 **From the notes (page 5), verbatim idea:** *"All the results are bound to this group and not any other group. Hence group relative..."*

$$\boxed{A_i = \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G)}}$$

Why this is the natural answer to §9's credit-assignment problem, without PPO's extra value network: instead of *training a separate model* to predict a baseline value at every step (PPO's $V_\phi(s)$), GRPO gets its baseline "for free" by sampling several completions to the *same* prompt and using their own average as the baseline — a same-prompt group takes the place of a learned critic. This is a direct engineering trade-off: GRPO removes the cost and instability of training a value network, at the cost of needing to sample G completions per prompt during training rather than 1.

12. Full Worked Example: GRPO Advantages

Group of $G = 4$ sampled completions for one prompt; reward $+1$ for a correct final answer, -1 for incorrect (a common simple reward design for reasoning tasks, as in DeepSeek-R1-style training):

Group A rewards: $(1, 1, -1, 1)$ — 3 correct, 1 incorrect.

$$\text{mean} = 0.5, \qquad \text{std} = 0.866$$

$$A = \left(\frac{1-0.5}{0.866}, \frac{1-0.5}{0.866}, \frac{-1-0.5}{0.866}, \frac{1-0.5}{0.866} \right) = (0.577,\ 0.577,\ -1.732,\ 0.577)$$

Group B rewards (same prompt structure, different group — e.g. a harder prompt where the model rarely succeeds): $(-1, -1, -1, 1)$ — 1 correct, 3 incorrect.

$$\text{mean} = -0.5, \qquad \text{std} = 0.866, \qquad A = (-0.577, -0.577, -0.577,\ 1.732)$$

The key structural point, directly illustrating page 5's "bound to this group" note: the lone correct completion in Group B receives a **large positive** advantage ($+1.732$) for succeeding on a prompt where success was rare, while a correct completion in Group A receives a smaller positive advantage ($+0.577$), because success was common there. **Absolute reward alone (+1 in both cases) does not distinguish these two situations; the group-relative advantage does** — this is precisely the sense in which GRPO's normalization is informative and not just a cosmetic rescaling.

13. Exam Quick-Reference Sheet

Concept	Formula / Definition
Bellman equation	$V^\pi(s) = \sum_a \pi(a\|s) \sum_{s'} P(s'\|s,a)[R + \gamma V^\pi(s')]$
Bellman optimality	$V^*(s) = \max_a \sum_{s'} P(s'\|s,a)[R + \gamma V^*(s')]$
Advantage	$A(s,a) = Q(s,a) - V(s)$
Policy gradient	$\nabla_\theta J = \mathbb{E}[\nabla_\theta \log \pi_\theta(a\|s) \cdot A(s,a)]$
PPO clipped objective	$L^{\text{CLIP}} = \mathbb{E}[\min(r_t A_t,\ \text{clip}(r_t, 1-\epsilon, 1+\epsilon) A_t)]$
GRPO advantage	$A_i = \dfrac{r_i - \text{mean}(\text{group})}{\text{std}(\text{group})}$
Reward hacking	Optimizing the literal reward signal in a way that violates designer intent

Common Exam Traps

- **"An agent that reward-hacks is malfunctioning/buggy"** — false; per §6, it is correctly optimizing exactly the objective it was given — the failure is in reward *specification*, not agent behavior.
- **"PPO and GRPO solve different problems"** — false-ish/imprecise; both are answers to the *same* credit-assignment problem (§9) — PPO via a learned value baseline, GRPO via a sampled-group baseline (§10–11).

- **"GRPO needs a critic/value network like PPO"** — false; that's the entire point of the group-relative normalization — §11.
- **"Discount factor γ only matters for infinite-horizon problems"** — false; even the 2-step MDP of §4 uses γ, and it matters whenever reward is delayed at all, finite horizon or not.
- **"Higher reward always means better group-relative advantage in GRPO"** — false; §12 shows a $+1$ reward can yield a *small* advantage in an easy group or a *large* one in a hard group — it's entirely relative to the group.

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14. Practice Exercises

Q1. A 4-state deterministic chain $1 \to 2 \to 3 \to 4$ (terminal) has rewards $R(1{\to}2) = -1$, $R(2{\to}3) = -1$, $R(3{\to}4) = +20$, and $\gamma = 0.9$. Compute $V(1), V(2), V(3), V(4)$ by back-substitution, showing every step.

Q2. A robot's reward function is "+1 point per second the room is measured as clean." Using the vacuum-robot example from §6 as a template: (a) propose one additional way this specific reward function could be gamed, distinct from the dirty-then-clean exploit already described. (b) Propose a redesigned reward function (§7) that is more robust to your proposed exploit, and briefly argue why.

Q3. For a group of $G = 5$ GRPO-sampled completions with rewards $(1, 1, 1, 1, -1)$: (a) compute the group mean, standard deviation, and each completion's advantage. (b) A second group (different, harder prompt) has rewards $(-1, -1, -1, -1, 1)$. Compute its advantages too. (c) Compare the advantage assigned to the single correct completion in each group, and explain the difference in one sentence connecting to §12's key point.

Q4. In one paragraph, explain why RLHF's credit-assignment problem (§9) is structurally the same difficulty PPO and GRPO were both designed to address in classical RL (§8, delayed/sparse terminal reward), even though the "state" in an LLM setting is a partial text sequence rather than, e.g., a robot's physical position.

Q5. Using the PPO clipped objective (§10) with $\epsilon = 0.2$: for advantage $A_t = -0.8$ (a bad action) and ratio $r_t = 0.75$ (policy wants to *decrease* this action's probability, consistent with the negative advantage), compute both the raw and clipped terms and the resulting L_t^{CLIP}. Does clipping activate here? Explain why decreasing probability for a bad action is generally *not* the case PPO's clip is designed to restrain, contrasting with the §10 worked example (where clipping restrained *increasing* probability for a good action).

Q6. [Design/synthesis, open-ended — graded on reasoning quality.] You are designing the reward function for an RLHF-trained customer-support chatbot, where the reward model is trained from human pairwise preference comparisons (§7.1). (a) Propose one concrete way this reward model could be gamed by the policy in a way that technically increases learned reward while making the chatbot objectively worse for users (in the spirit of §6). (b) Propose one guardrail (§7) to mitigate it. (c) Explain, referencing §7.1's lottery/utility-elicitation device, why training the reward model from pairwise comparisons rather than asking humans to output a raw numeric score directly is a reasonable design choice.

Solutions

Q1. $V(4) = 0$. $V(3) = 20 + 0.9(0) = 20.0$. $V(2) = -1 + 0.9(20.0) = 17.0$. $V(1) = -1 + 0.9(17.0) = 14.3$. $\boxed{V(1) = 14.3,\ V(2) = 17.0,\ V(3) = 20.0,\ V(4) = 0}$.

Q2. (a) E.g., the robot could position an object or its own sensor to *misreport* cleanliness (e.g., hover its sensor over an already-clean patch, or cover a dirty patch with an object) rather than actually cleaning — gaming the *measurement* rather than re-creating dirt, a distinct exploit from the notes' dirty-then-clean cycle, since here the ground truth never changes but the reported/measured signal does. (b) A more robust design would reward **reduction in verified total dirt over an episode**, capped once the room reaches a clean state (removing the incentive to re-dirty, since no further reward is available once clean), combined with an independent verification/audit step that can't be manipulated by the same actuator that's being rewarded (addressing the sensor-gaming exploit from (a)) — directly applying §7's "constraints and guardrails" principle to a redesigned, harder-to-game specification.

Q3. (a) Mean $= 0.6$, std $= 0.8$. Advantages: correct completions $\frac{1-0.6}{0.8} = 0.5$ each (×4); incorrect: $\frac{-1-0.6}{0.8} = -2.0$. (b) Mean $= -0.6$, std $= 0.8$. Advantages: incorrect $\frac{-1-(-0.6)}{0.8} = -0.5$ each (×4); the single correct: $\frac{1-(-0.6)}{0.8} = 2.0$. (c) The lone correct completion in the *hard* group (b) gets advantage $+2.0$, four times larger than a correct completion's advantage of $+0.5$ in the *easy* group (a) — directly matching §12: identical raw reward ($+1$) is worth far more, in group-relative terms, when it represents a rare success against a group that mostly failed, than when success was already the norm.

Q4. In classical RL, an episode consists of many state transitions, and if reward only arrives at the very end (e.g., win/lose a game), the agent must somehow figure out which of its many earlier actions actually deserved credit for that final outcome — the classical credit-assignment problem PPO addresses via a learned value baseline at every intermediate state. In RLHF, a "state" is a partial generated sequence (the tokens produced so far) and the "episode" is one full generation; the reward (e.g., a human/reward-model judgment of overall response quality) is typically available only once the full response is complete, exactly mirroring the classical case: many token-level decisions, one terminal reward, and no direct observation of how good any individual token choice was in isolation. The mechanism differs (text tokens vs. physical actions) but the structural problem — many sequential decisions, one delayed evaluative signal — and the class of solutions (learned baselines/PPO, or group-relative baselines/GRPO) are the same.

Q5. Raw term: $r_t A_t = 0.75 \times (-0.8) = -0.6$. Clipped ratio: $\text{clip}(0.75, 0.8, 1.2) = 0.8$ (0.75 is below the lower bound 0.8, so it's clamped up to 0.8). Clipped term: $0.8 \times (-0.8) = -0.64$. $L_t^{\text{CLIP}} = \min(-0.6, -0.64) = -0.64$. Clipping **does** activate here, but note the direction: because $A_t < 0$, the objective is more negative (worse) at the *clipped* value than at the raw ratio — the min operator here is selecting the clipped (more pessimistic) term, which in this configuration doesn't restrain the update the same protective way as the §10 example. The general point: for a bad action, PPO *wants* the policy to decrease r_t further (that's the gradient direction that improves the objective, since $A_t < 0$); the clip's protective role is specifically to prevent the objective from rewarding *pushing r_t too far in the improving direction* — here, that means clip protects against decreasing r_t *too much below* $1 - \epsilon = 0.8$, not against increasing it — so, unlike the good-action case where clipping capped an over-eager probability increase, here it caps an over-eager probability *decrease*, the mirror-image restraint for the opposite-signed advantage.

Q6. Sample strong answer: (a) The policy could learn to produce long, verbose, superficially agreeable, or hedge-everything responses that human raters tend to slightly prefer in isolated pairwise comparisons (a well-documented RLHF failure mode: reward models can be biased toward length/sycophancy), technically raising the learned reward while making the chatbot less useful/more evasive for actual

customer problems — a direct analogue of the vacuum robot exploiting its literal reward signal rather than the designer's true intent (§6). (b) A guardrail: incorporate an explicit penalty or auxiliary constraint against excessive length/hedging relative to a reference, or mix in reward signal from actual task-completion metrics (e.g., did the customer's issue get resolved) rather than relying solely on pairwise preference-model score, per §7's "proper constraints and guardrails" principle. (c) Per §7.1, the lottery device works because humans are often much better at *comparing* two options than at assigning a stable, consistent absolute number to either one in isolation — asking "is this a 7 or an 8 out of 10" is noisy and hard to calibrate across raters, while "which of these two responses is better" is a much easier, more consistent judgment to make; pairwise comparison training (fitting a reward model whose implied ordering matches these comparisons) is the large-scale, learned version of exactly this elicitation principle, trading the difficulty of eliciting absolute utility for the easier task of eliciting relative preference.

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